Lab 1 — Teacher Notes

Lab: CS-2007L Week 1 · Introduction to C# Environment & Basic Data Organization
Duration: 3 hours
Decks: `slides/A_lecture.md` (project) · `B_teacher-understanding.md` (prep) · `C_viva-bank.md` (viva)
Script: `04_lecture-script-roman-urdu.md`

Before the lab (night before)

1. Read Deck B completely (~30–40 min).
2. Rehearse Example 2 debugger path once on your machine.
3. Confirm `.NET 8` on demo PC: `dotnet --version`.
4. Print or pin rubric (2 + 3 + 2 + 3 = 10).
5. Decide AI policy for this week (allowed with disclosure / not allowed) and announce once.

Materials to open

Order	File
1	Deck B (private)
2	Deck A + Roman Urdu script
3	Lab manual / examples
4	Deck C for last 5 minutes

3-hour timeline (with buffers)

Clock	Block	Deck A slides	Notes
0:00–0:15	Hook + outcomes	1–3	Keep short; energy high
0:15–0:35	Toolchain live	4–8	Every student must `dotnet run` once
0:35–0:55	Debugger	9–11	You drive; they follow
0:55–1:15	Types + records	12–15	Board drawing for heap
1:15–1:35	Six ops	16–18	Emphasize insert O(n)
1:35–1:50	Tradeoff	19–20	Release build reminder
1:50–2:40	Tasks 1–4	21	Circulate; Task 4 = debugger
2:40–2:55	Wrap / homework	22	Tasks 5–6 + Q1–Q3
2:55–3:00	Exit ticket	Deck C	1–2 random Qs

Circulation priorities during Tasks 1–4

1. Students stuck on `dotnet` PATH
2. Task 4 (force debugger use — not "guess the bug")
3. Task 2 `TryParse` loop

4. Task 3 written parts (don't let them skip)

Marking shortcuts

- **Excellent demo:** builds + shows breakpoint confidently
- **Satisfactory:** runs with prompts
- **Needs improvement:** cannot build or never opens debugger

AI disclosure missing when AI was used → CLO-3 marks suffer.

Pitfalls this week

- Top-level statements vs `Main` confusion → insist `--use-program-main` / "Do not use top-level statements"
- Students pasting all examples without reading Observe notes
- Example 6 timed in Debug → weird numbers
- Task 5 early use of `List<T>` — redirect to arrays

Homework announcement (exact)

"Assignment block Labs 1–4 → Assignment 1. Lab 1 post-lab Q1–Q3 is part of that. Quiz 1 covers Labs 1–3."

After lab

- Note who only finished Tasks 1–2 (follow up)
- Save any good student questions for next year's Deck B
- Prep Lab 2: frequency count + Big-O reading from theory lectures 4–6